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# Softmax Temperature, 50% Thresholds, and Q-Value Differences

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## Introduction: softmax temperature and inverse temperature in reinforcement learning policies

*In a softmax policy, temperature controls how random versus greedy action choice is. High temperature makes probabilities flatter and closer to uniform, while low temperature makes the policy concentrate on the highest-Q action. (28 sources)*

In softmax reinforcement learning policies, action probabilities are computed from the current action values Q(s,a) by exponentiating them and normalizing, so actions with larger Q-values get larger choice probabilities while preserving their rank order. The main control knob is the **temperature** T or \tau, which sets how strongly differences in Q-values affect choice; many papers describe this as controlling the amount of randomness or the exploration–exploitation tradeoff. (Moriyama et al., 2007) (Schaul et al., 2019) (Ongchoco et al., 2024) (Khalid et al., 2025)

Across the literature, the interpretation is very consistent: **higher temperature** makes the policy flatter and closer to random or uniform choice, which increases exploration, whereas **lower temperature** makes the policy more concentrated on the best-valued action, which increases exploitation. In the low-temperature limit, the policy approaches greedy argmax selection; in the high-temperature limit, all actions become nearly equally likely. (Nocera et al., 2014) (Reverdy et al., 2015) (Perez-Romero et al., 2016) (Barsce et al., 2017) (Smith et al., 2020) (Wang et al., 2020) (Meder et al., 2020) (Amin et al., 2021) (Condorelli et al., 2023) (Wang et al., 2024) (Lomis et al., 2025)

Some papers use the equivalent **inverse temperature** form, often written as \beta, \alpha, \eta, \lambda, w, or similar. This simply flips the direction of the parameter: larger inverse temperature means a steeper, more value-sensitive, more greedy policy, while inverse temperature near zero gives nearly random choice. (Ruiz-Serra et al., 2023) (Ger et al., 2024) (Harrell et al., 2025) (Higashi, 2025) (Nair, 2025) (Sultana et al., 2026) (Ki et al., 2026)

A useful way to think about temperature is that it changes the **softness** or **steepness** of the mapping from values to probabilities, not the ordering of actions itself: low temperature sharpens small Q-value differences into big probability differences, and high temperature smooths them out. This same idea appears in reinforcement learning, policy distillation, and entropy-regularized control. (Gevaert et al., 2022) (Liu et al., 2020) (Wu et al., 2018) (Lotzsch, 2018) (He et al., 2025) (Chen et al., 2026)

### Evidence

**(Moriyama et al., 2007)**

Reinforcement Learning on a Futures Market Simulator

*"Softmax calculates action choice probabilities p s t (a), where a ∈ A(s t ), as"*

*"T is called the temperature, with T > 0, and controls the effect of randomness."*

**(Schaul et al., 2019)**

Adapting Behaviour for Learning Progress

*"Temperature: a Boltzmann softmax policy based on action-logits, modulated by temperature, T ."*

**(Ongchoco et al., 2024)**

When New Experience Leads to New Knowledge: A Computational Framework for Formalizing Epistemically Transformative Experiences

*"A SoftMax function converts a vector of real numbers into a vector of probabilities in such a way that preserves the ordinality of the original numbers (i.e., if v = (a, b) and SoftMax(v) = (p a , p b ), then p a > p b if and only if a > b). Thus, under a SoftMax policy, actions with higher expected values are more likely to be selected than actions with lower expected values. Importantly, the softMax function includes a temperature parameter (call it T ) which controls how deterministic the policy is. For very small values of T, the policy is nearly deterministic, meaning that the agent will almost always choose the action with the highest expected value. When T is very large, the policy is nearly uniformly random, so that each action is equally likely to be selected. While the exact value of T is not important for the analysis that follows"*

*"the probability associated with each action is directly correlated to the expected value of that action (given the belief state)."*

**(Khalid et al., 2025)**

Adaptive Traffic Lights Control System Using Deep Q-Network and Salp Swarm Optimization

*"Softmax exploration offers an alternative by assigning probabilities to actions according to their Q-values and a temperature parameter, allowing for a smoother balance between exploration and exploitation as the temperature decreases over time."*

**(Nocera et al., 2014)**

The role of intrinsic motivations in attention allocation and shifting

*"Here, the temperature τ controls the exploration strategy: the higher the temperature, the closer the strategy is to a random policy (exploration); the lower the temperature, the closer the strategy is to Q(s, a) maximization (exploitation)."*

**(Reverdy et al., 2015)**

Parameter Estimation in Softmax Decision-Making Models With Linear Objective Functions

*"The degree of stochasticity is controlled by the temperature \tau . In the limit \tau \rightarrow 0 + , the rule reduces to the standard maximum and deterministically selects the option with the highest value of V i . In the limit \tau \rightarrow +\infty, all options are equally probable and the rule selects options according to a uniform distribution."*

**(Perez-Romero et al., 2016)**

Power-Efficient Resource Allocation in a Heterogeneous Network With Cellular and D2D Capabilities

*"where τ is the so-called temperature parameter. High temperature causes the different options to be nearly equiprobable.*

*In contrast, low temperature leads to a greater difference in selection probability for APs/BSs that differ in their Q value estimates, and the higher the value of Q, the higher the probability of selecting the corresponding AP/BS."*

**(Barsce et al., 2017)**

Towards autonomous reinforcement learning: Automatic setting of hyper-parameters using Bayesian optimization

*"The most common implementation of this algorithm is the Boltzmann Softmax, which defines the probability of taking the action a in state s as"*

*"where τ is the computational temperature. By using this policy, the agent starts by employing an exploratory policy at the beginning with a"high temperature", switching over time to a more greedy policy as the value of the hyper-parameter τ is increasingly reduced, i.e. exploration "cools down"."*

**(Smith et al., 2020)**

Learning to play against any mixture of opponents

*"The temperature parameter allows us to control the softness of the maximum operator. A high temperature produces actions that have a near-uniform probability, and as the temperature is lowered the distribution concentrates weight on the highest Q-Values (Sutton & Barto, 2018)."*

**(Wang et al., 2020)**

Reannealing of Decaying Exploration Based On Heuristic Measure in Deep Q-Network

*"Note that hyperparameter T controls the exploration (Ishii et al., 2002). If the temperature is high, the action selection according to π approaches uniform distribution, which yields more randomness and thus encourages exploration. On the other hand, low temperature would reduce the randomness and enhance exploitation. As an extreme case, if the temperature is zero, the negative entropy term in Eq. ( 4) goes away, and the corresponding policy becomes deterministic which takes the greedy action arg max π Q(s, a)π(a|s)da given the current estimate of Q(s, a). With the softmax probability, the possibilities for each action to be selected are ranked and weighted relevant to their estimated Q-values, instead of equal probabilities for all actions in ε-greedy approach."*

**(Meder et al., 2020)**

Development of directed and random exploration in children.

*"Importantly, the softmax choice contains a temperature parameter that governs the amount of randomness in the choice probabilities. This enables us to quantify the amount of random exploration for each learner. Higher temperature sampling corresponds to noisier predictions, where as □ □ □ ,all options have an equal probability of being chosen. Conversely, lower temperatures produce choice probabilities that are more concentrated on highvalue options, where as □ □ 0, it becomes an argmax choice rule (i.e., always choosing the option with the highest value)."*

**(Amin et al., 2021)**

A Survey of Exploration Methods in Reinforcement Learning

*"where T m > 0 is called the temperature and controls how frequent the agent will choose random actions as opposed to the best actions a t = arg max a Q t (s, a). If T m decreases, the probability of generating the action with the highest expected reward a t increases, leading to a decrease in the probability of selecting other actions and hence a lower probability of exploring the environment. In the limit of zero temperature T m □ 0, the agent uses the target policy to select greedy actions in order to maximize the obtained rewards."*

**(Condorelli et al., 2023)**

Cheap Talking Algorithms

*"The parameter τ t , called temperature, modulates the intensity of exploration: for smaller values of τ t , the probability mass increasingly concentrates on the action(s) that are most rewarding according to the current estimate Q t (s, a)."*

**(Wang et al., 2024)**

APPA-3D: an autonomous 3D path planning algorithm for UAVs in unknown complex environments

*"The action assignment for the Softmax distribution strategy is based on the Boltzmann distribution, which is represented by Eq. ( 18):"*

*"In Eq. ( 18), Q s, a j records the average reward of the current action; τ > 0 is called "temperature", The smaller of τ , the higher the probability of selecting actions with higher average rewards. When τ tends to 0, Softmax tends to " exploitation only", when τ tends to infinity. Softmax tends to "exploration only"."*

**(Lomis et al., 2025)**

Strategic Scenario Interaction: A Computational Framework Based on Game-Theoretic and Quantum-Inspired Modeling

*"Parameter T (temperature) regulates exploration vs. exploitation. As T □ 0 , agents behave deterministically; as T □ ∞ , their behavior approximates random choice"*

*".The Softmax rule translates the relative utilities of agents into choice probabilities through smooth logistic mapping. Higher utility values yield higher selection probabilities, whereas the temperature parameter T controls the degree of randomness in decisionmaking, from near-deterministic choices at low T to exploratory behavior at high T"*

*".Therefore the "collapse" is not triggered by any probability threshold (e.g., P > X) but occurs naturally at each iteration when an agent's probabilistic state is resolved into a concrete scenario through sampling."*

**(Ruiz-Serra et al., 2023)**

Inverse Reinforcement Learning as the Algorithmic Basis for Theory of Mind: Current Methods and Open Problems

*"The hyperparameter α serves as an inverse temperature parameter defining the steepness of the policy distribution, or how "greedy" for optimal Q-values it is."*

**(Ger et al., 2024)**

Using recurrent neural network to estimate irreducible stochasticity in human choice behavior

*"This parameter controls the level of stochasticity in the action selection, where β  0 corresponds to random choices and β  ∞ for deterministically choosing the highest value action."*

**(Harrell et al., 2025)**

A Foraging-Theory-Based Model Captures the Spectrum of Human Behavioral Diversity in Sequential Decision Making

*"β ( R + ) (aka "inverse temperature") is commonly thought of as controlling the amo noise in the decision process and therefore determines its stochasticity; β also determines the sensitivity to differences. When β is zero, the decision process is insensitive to value difference, completely random, an chooses uniformly between each option. As β approaches positive infinity, the SoftMax output becomes "greedy": it always selects the option with the highest value."*

**(Higashi, 2025)**

Individuality transfer: Predicting human decision-making across task conditions

*"The probability of selecting action a was determined by a softmax policy:"*

*"where q it > 0 was a parameter called the inverse temperature or reward sensitivity, controlling the balance between exploration and exploitation."*

**(Nair, 2025)**

Softmax is 1/2-Lipschitz: A tight bound across all ℓp norms

*"where Q(s, a) are the action logits for state s  S, A is the action space, and λ > 0 is the inverse temperature coefficient."*

**(Sultana et al., 2026)**

A Multi-Model Adaptive Q-Learning Framework for Robust Portfolio Management in Stochastic Markets

*"The beta parameter controls the "temperature" of the softmax function, which in turn controls the trade-off between exploration and exploitation. Higher values of beta create more deterministic (exploitation-oriented) behavior, and lower values of beta promote more exploration"*

*".The β parameter of softmax determines how action probabilities are weighted by their Q-values, modulating the explorationexploitation tradeoff."*

**(Ki et al., 2026)**

Direct Soft-Policy Sampling via Langevin Dynamics

*"In soft policy frameworks, a temperature parameter is commonly introduced to balance value maximization and entropy regularization. Specifically, the soft policy is defined as π  exp (wQ), where w > 0 controls the trade-off between expected return and policy entropy"*

*".To analyze the effect of the temperature parameter w"*

*"As w increases, the target Boltzmann distribution assigns higher probability mass to high-reward regions, leading to sharper concentration around the highest-reward modes."*

**(Gevaert et al., 2022)**

Distilling Deep RL Models Into Interpretable Neuro-Fuzzy Systems

*"Note that, if the temperature τ = 1, we obtain the original softmax function. If τ > 1, the result of this function is a smoothed version of the normal softmax"*

*". Note that the output of a softmax function on a vector of Q-values is usually a very smooth probability distribution, as Q-values often have very small yet important differences. For this reason, the temperature τ in policy distillation is usually lower than 1, resulting in sharper probability distributions."*

**(Liu et al., 2020)**

Developing Multi-Task Recommendations with Long-Term Rewards via Policy Distilled Reinforcement Learning

*"The temperature τ controls how much knowledge is transferred from a teacher to the student. Raising the temperature will soften the transformed probability distribution and allow more knowledge to be transferred to the student [63]."*

**(Wu et al., 2018)**

Inverse POMDP: Inferring What You Think from What You Do

*"When the policy is optimal, the term P θ (a t |b t = i, o t ) is a delta function, so the derivative of the policy does not exist. As a result,*

*we approximate the optimal policy using a softmax or Boltzmann policy with a small learnable temperature τ . The softmax introduces an additional sub-optimality of the agent: instead of choosing the action that brings the maximal expected reward, the agent has some chance of choosing a reward that yields a lesser reward, depending on the state-action value Q. Under the softmax policy, the actions under belief state b follow the distribution"*

**(Lotzsch, 2018)**

Using Deep Reinforcement Learning for the Continuous Control of Robotic Arms

*"The parameter is called the temperature and regularizes the amount of randomization induced. When decreases, the policy becomes more and more deterministic or greedy, while increasing encourages exploration."*

**(He et al., 2025)**

Random Policy Valuation is Enough for LLM Reasoning with Verifiable Rewards

*"we transition from deterministic to stochastic action selection by converting Q πu into a soft sampler, i.e., π s (a|s"*

*", where ρ is a temperature parameter. This strategy selects actions proportional to their estimated success probability, which is able to explore multiple reasoning pathways for improving diversity, rather than committing to a single path"*

*".Theorem 3.2 characterizes that the temperature ρ trades off between diversity and quality."*

**(Chen et al., 2026)**

Safe Hierarchical Autonomous Driving Framework: Integrating SAC-D Planning With HOCLF-HOCBF-QP Constrained Low-Level Control

*"For better balance exploration and exploitation, the temperature hyperparameter α can be automatically adjusted by using the following objective functions with target entropy constant  :"*

*"Then, a softmax operation is applied to obtain the action probabilities"*

*".The actor and critic interact through an entropy-regularized Bellman backup, in which the target state value is computed as the expectation over the policy of the minimum target Q-value minus an entropy term scaled by the temperature parameter. The policy is optimized to maximize expected returns while encouraging exploration via entropy regularization. The entropy temperature is automatically tuned during training to match a target entropy proportional to the size of the discrete action space, enabling adaptive exploration throughout learning."*

# 50% choice probability threshold and Q-value crossover

*In a two-action softmax, the 50% choice point stays at the same place: it happens when the two action values are equal. Temperature changes how steeply probability moves around that point, but with a fixed common temperature it does not move the Q-value crossover itself. (9 sources)*

For **two actions**, softmax reduces to a **logistic choice rule** in the **value difference** Q(A)-Q(B), so the natural crossover is at **zero value difference**: if Q(A)=Q(B), then each action has probability 0.5. What temperature or inverse temperature changes is the **slope** around that point, not its location. Inverse temperature controls how sensitive choice is to the value difference, with small values giving a shallow curve and more random responding, and large values giving a steep curve and more consistent, near-greedy choice. (Zhang et al., 2019) (Katahira et al., 2022) (Woo et al., 2025) (Findling et al., 2024) (Xu et al., 2026)

This means that with a **fixed temperature shared across actions**, changing temperature does **not shift the Q-value point where choice flips from below 50% to above 50%**; the 50% threshold remains at Q(A)-Q(B)=0. Instead, low inverse temperature or high temperature makes probabilities stay closer to 0.5 when values are close, while high inverse temperature makes even small value gaps produce strong preferences. This is exactly the pattern described in two-choice examples where high temperature gives probabilities near 0.5 for each action and low temperature strongly favors the higher-valued action even when beliefs are close. (Duffin et al., 2014) (Zhang et al., 2019)

Some models also add other terms, such as a **stay bonus**, noise, or bias term, on top of the Q-values before the softmax. In those cases, the 50% point can shift because the effective decision variable is no longer just Q(A)-Q(B). So the “no shift” statement applies to the plain two-action softmax with one common temperature, not to models with extra additive choice biases. (Morris et al., 2019) (Molinaro et al., 2025)

A related practical point is that some studies **fix** inverse temperature at a large value for fitting or identifiability reasons. Fixing \beta changes how strongly value differences map into probabilities, but it still does not by itself move the equal-probability crossover in the standard two-action softmax. (Weydmann et al., 2022) (Molinaro et al., 2025)

## Evidence

### (Zhang et al., 2019)

Using reinforcement learning models in social neuroscience: frameworks, pitfalls and suggestions of best practices

> *"In the case of two choice options, the Softmax function is simplified as a logistic curve, where the input is the value difference between V(A) and V(B), and the output is the probability of choosing A (equation 3). The inverse temperature parameter (τ ) is the slope of the sigmoid curve, which measures choice consistency. When τ is small (e.g. τ = 0.2), the curve is shallow, which represents more random choices (i.e. less consistent)"*

> *".In contrast, when τ is large (e.g. τ = 5), the curve is steep, and this represents more consistent choices, in favor of the high(er) reward option (Figure 1C)."*

### (Katahira et al., 2022)

Influences of Reinforcement and Choice Histories on Choice Behavior in Actor-Critic Learning

> *"the inverse tem- perature parameter  [0] ∞) controls how sensitive the choice probabilities are to the value difference between actions"*

### (Woo et al., 2025)

An Information-Theoretic Framework for Understanding Learning and Choice Under Uncertainty

> *"the difference between these values is used to make a choice (Figure 2a)"*

> *".On each trial, the choice is determined stochastically based on the probability of selecting each option, P j , computed using the softmax function, as follows:*

> *where the indices j and k correspond to the two choice alternatives, and β is the inverse temperature parameter that controls the sensitivity to value differences (i.e., slope of the softmax function)."*

### (Findling et al., 2024)

Computation noise promotes zero-shot adaptation to uncertainty during decision-making in artificial neural networks

> *"As in existing theories, we modeled the choice process using a stochastic softmax action selection policy, controlled by an inverse temperature β where B(. ) denotes the Bernoulli distribution and Q t,A and Q t,B correspond to the values associated with actions A and B, respectively. This stochastic action selection policy reduces to a purely greedy (value-maximizing) "argmax" policy when β  ∞."*

### (Xu et al., 2026)

Reinforcement Learning Measurement Model

> *"where the inverse-temperature parameter β ≥ 0 quantifies the extent to which higher-valued actions are preferred over lower-valued ones. Larger values of β correspond to more consistent, value-aligned choices, whereas values near zero approximate random responding."*

### (Duffin et al., 2014)

Differential effects of reward and punishment in decision making under uncertainty: a computational study

> *"A low temperature gives a high probability of choosing the action with the highest belief, even when the beliefs are quite close. A high temperature results in mainly random responses, giving probabilities close to 0.5 for each of the actions, for the models which have only two actions"*

> *". As there were only two possible actions, the probability of making a type 2 response was given by 1 − P(t)."*

### (Morris et al., 2019)

Model-Free RL or Action Sequences?

> *"After combining the model-based and model-free Q-values, agents chose actions with probability proportional to the exponent of*

*the combined Q-values (plus a "stay bonus" capturing the tendency to repeat previous actions5 ). Formally, the probability of choosing action a in state s was given by a softmax function with inverse temperature parameter β, with a stay bonus v:"*

**(Molinaro et al., 2025)**

Reward function compression facilitates goal-dependent reinforcement learning

*"Q values were converted to a policy via the softmax function (Luce, 1959) with inverse temperature β and the noise parameter  to capture random errors in the participants' choices:"*

*"Because it trades off with other parameters (e.g., the learning rate -potentially undermining parameter recoverability), β = 25 was a fixed parameter in all models, as is common practice with similar models and tasks [14](Westbrook et al., 2024)."*

**(Weydmann et al., 2022)**

Switching to online: Testing the validity of supervised remote testing for online reinforcement learning experiments

*"The rule for choosing actions in response to a stimulus was defined stochastically by a Softmax Choice Policy: P(a|s) = exp(βQ(s, a))/Σ i (exp (βQ(s, a))), where β is the inverse temperature parameter which determines the degree to which differences in Q-values are translated into a more deterministic choice; the sum is over the n A = 3 possible actions"*

*"Based on previous studies using the RLWM task (Masters et al., 2020), we decided to fix β as 50 because parameters distribution was close to normal in this way and because free beta ranges (e.g., 0 to 50 or 0 to 500) lead to extreme values of beta in many cases with minimal or no change on function cost."*

# Choice probabilities versus Q-value differences in two-action versus multi-action settings

*With two actions, choice probability is a direct sigmoid of the single value difference \(Q(A)-Q(B)\), so one probability fully determines the other. With more than two actions, each action's probability depends on its value relative to all other actions, so there is no single universal mapping from one pairwise Q-difference to one choice probability. (10 sources)*

In the **two-action** case, softmax has a special simplification: it becomes a **logistic function of the value difference** between the two options. That means the probability of choosing A is fully determined by Q(A)-Q(B), and the probability of choosing B is just the complement, 1-P(A). Several papers state this directly, describing two-choice softmax as a sigmoid in the value difference and noting that inverse temperature sets the slope or sensitivity of that mapping. (Zhang et al., 2019) (Katahira et al., 2022) (Woo et al., 2025) Duffin et al. make the same point in practical terms: when there are only two actions, probabilities tend toward about 0.5/0.5 at high temperature, and one action's probability is simply one minus the other's. (Duffin et al., 2014)

This is why, in two-action tasks, people often talk about **choice probability as a function of the Q-value difference** rather than the raw Q-values themselves. As inverse temperature increases, that function gets steeper, so a small positive difference can already imply a high chance of choosing the better-valued option; as inverse temperature goes to zero, the policy becomes insensitive to the difference and approaches random choice. (Zhang et al., 2019) (Findling et al., 2024) (Harrell et al., 2025) (Xu et al., 2026)

In **multi-action** settings, the relationship is different. Softmax still preserves order—higher-Q actions get higher probabilities—but each action's probability is computed by normalizing over **all available actions**, not just one competitor. So P(a) depends on Q(a) relative to the whole set Q(i), and the same pairwise gap Q(A)-Q(B) can lead to different probabilities depending on the values of the other actions. (Weydmann et al., 2022) (Ongchoco et al., 2024) In other words, in a 3+-action softmax, there is generally **no single pairwise "50% threshold" story** like there is in the binary case, because probability mass is split across multiple options. (Model-Generated)

This also explains why statements like "high temperature gives probabilities close to 0.5" are specific to **two-action** models. With n actions, very high temperature pushes the policy toward **uniform choice over all actions**, so each action approaches probability 1/n, not 0.5. Papers using 3-action softmax write the policy explicitly as a normalization over all n_A=3 actions, and more general descriptions say that very high temperature makes each action nearly equally likely. (Weydmann et al., 2022) (Ongchoco et al., 2024) (Harrell et al., 2025)

So the cleanest summary is: **for two actions, probabilities map to the single Q-difference; for more than two, probabilities map to each action's Q-value relative to the full choice set**. Temperature changes how sharp that mapping is in both cases, but only in the binary case does it reduce to one sigmoid over one value difference. (Zhang et al., 2019) (Weydmann et al., 2022) (Sultana et al., 2026) (Ongchoco et al., 2024)

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